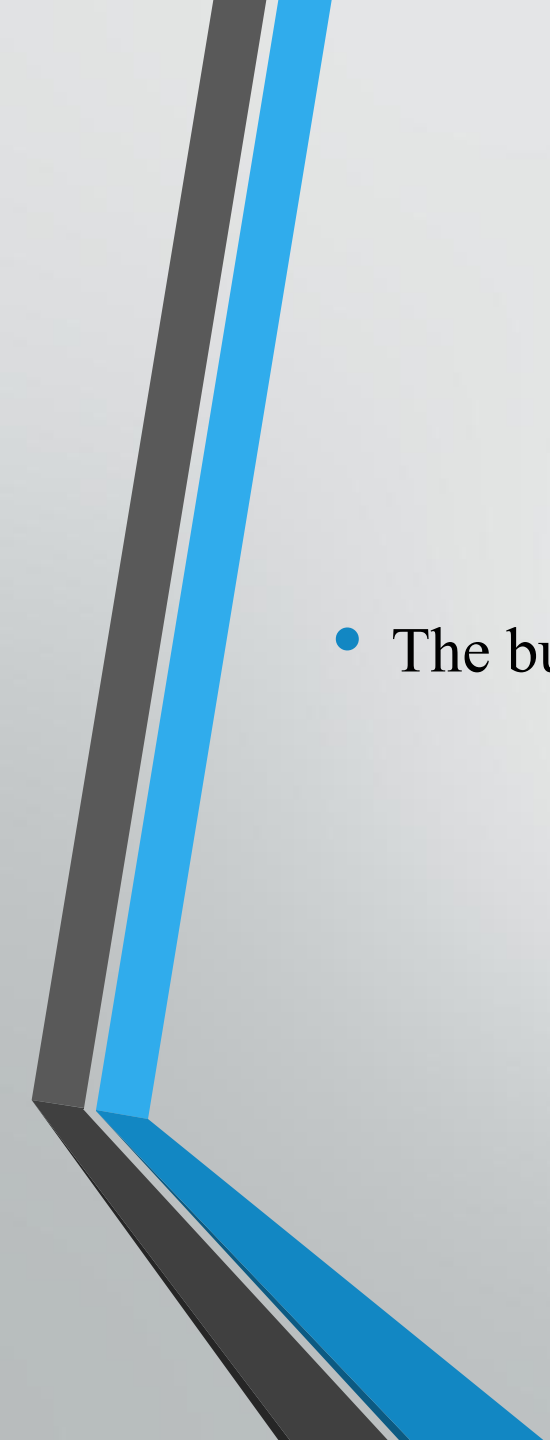




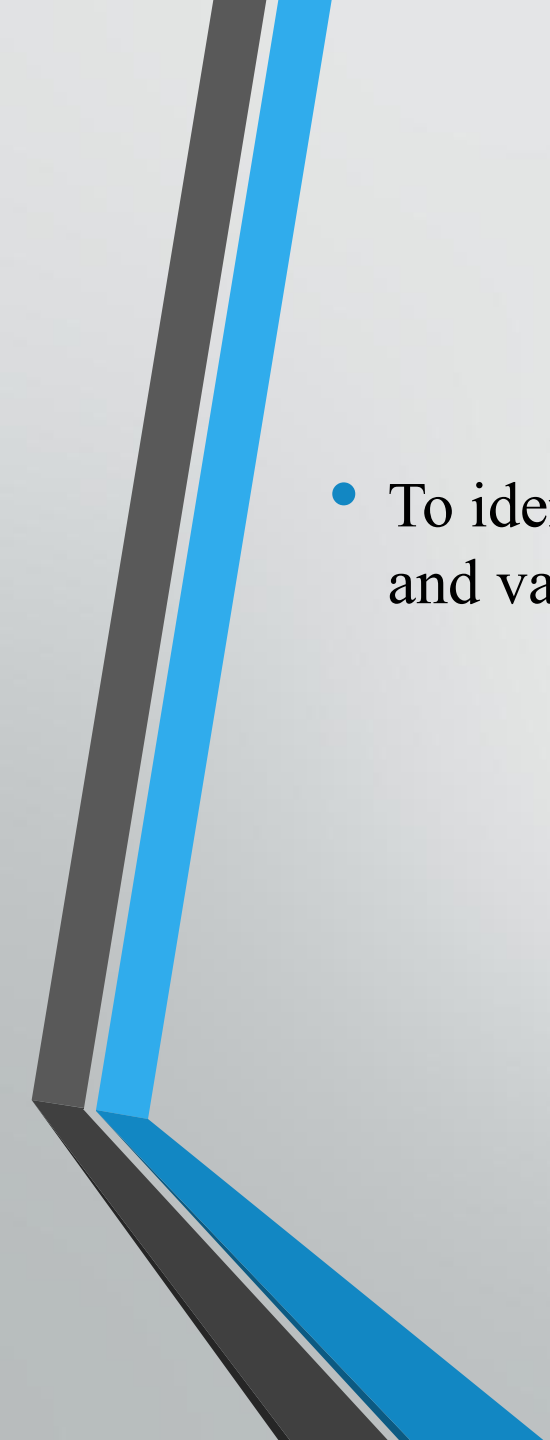
Sales & Profit Deep-Dive Analysis

Data Storytelling & Statistical
Validation



Problem Statement

- The business is generating high sales but still experiencing significant losses.



Objective

- To identify the root cause of negative profitability and validate it using statistical analysis.



Total Sales	Total Cost	Total Profit (loss)	Profit Margin	Avg Order Value
5.02M	63.84M	-58.82M	-1171.94%	5.02K

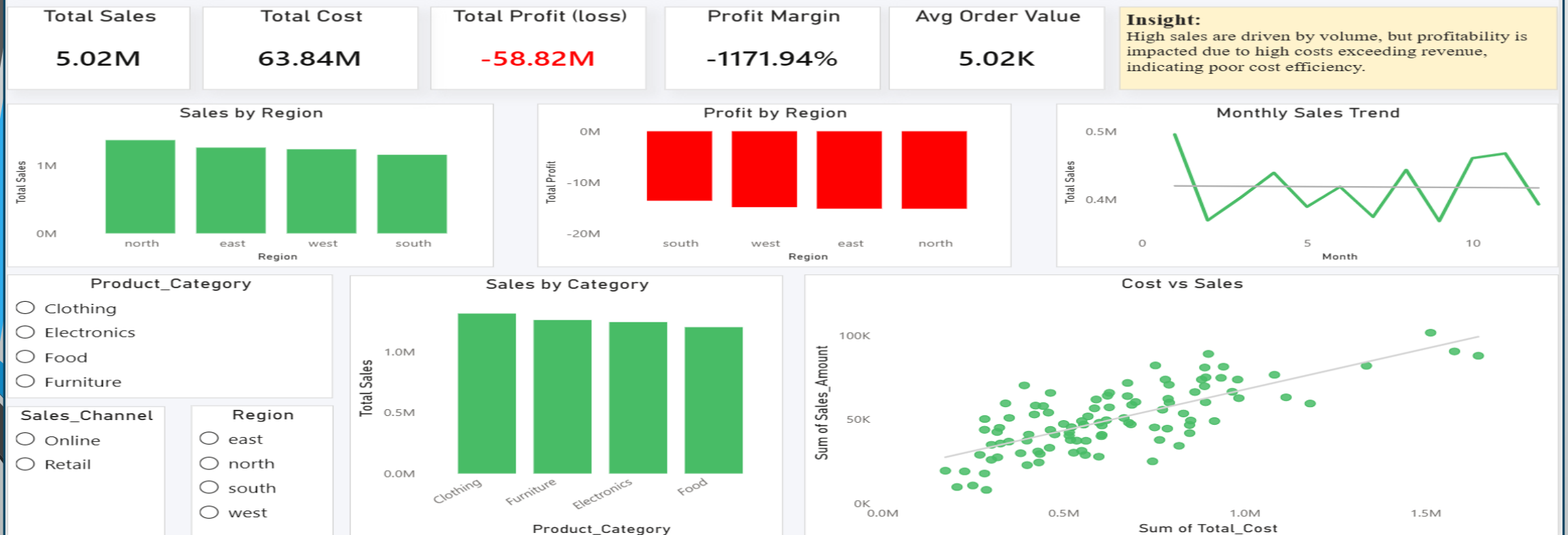
Key Metrics

- Total Sales: 5.02M
- Total Cost: 63.84M
- Total Profit: -58.82M
- Profit Margin: Negative

Key Observations

- Sales are high across all regions
- Profit is negative in all regions
- High revenue categories still show low profit
- Costs increase with sales

Sales & Profit Deep-Dive Analysis Dashboard



Profit Margin Analysis

- Average Profit Margin is negative
→ Most transactions are loss-making

```
df['Profit_Margin'] = df['Profit'] / df['Sales_Amount']  
df['Profit_Margin'].describe()
```

```
count    1000.000000  
mean      -31.168042  
std       102.461187  
min      -1808.297146  
25%       -23.460517  
50%        -9.477552  
75%        -2.931847  
max         0.990833  
Name: Profit_Margin, dtype: float64
```

Profit Margin Analysis:

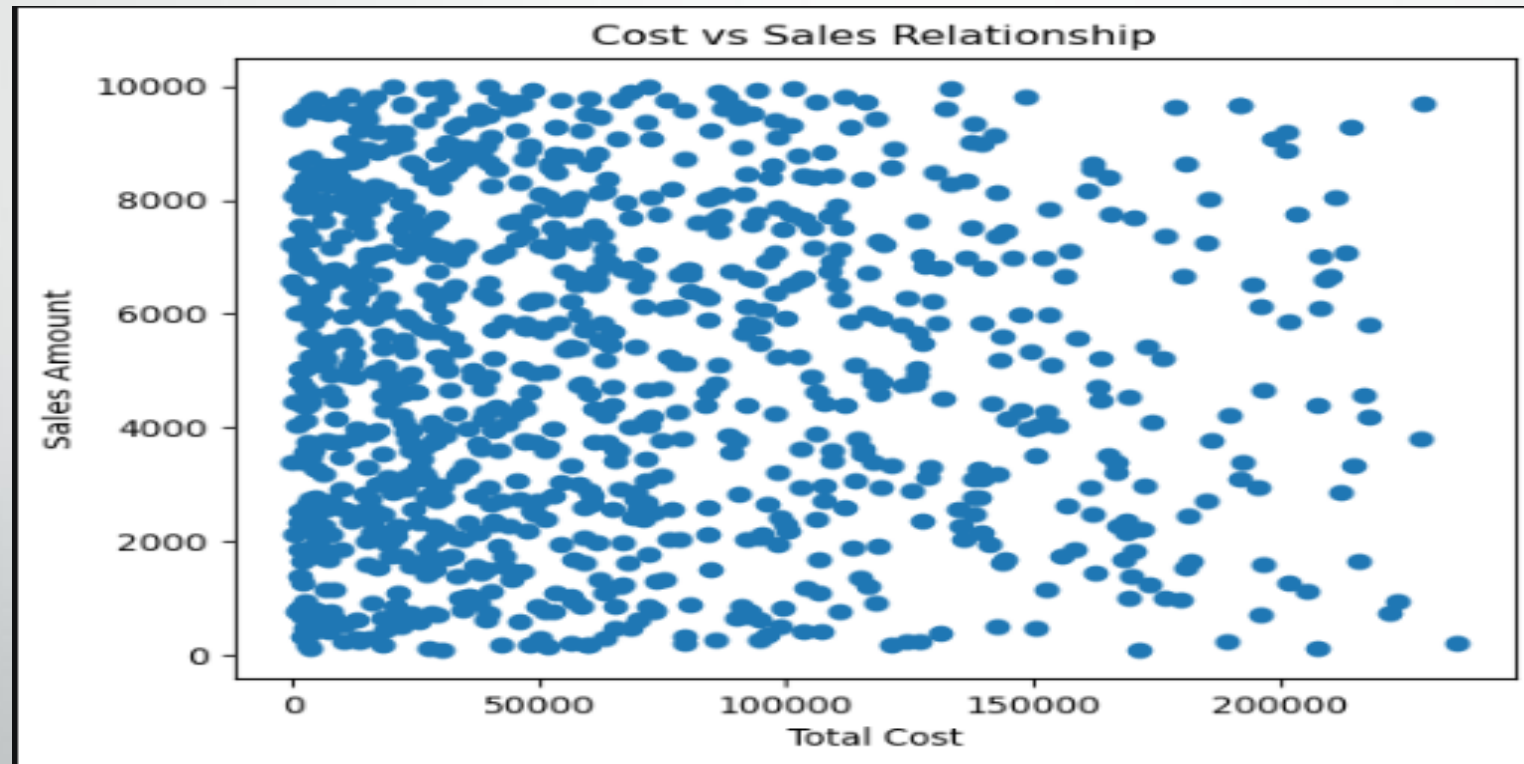
The average profit margin is negative, indicating that most transactions are not profitable.

Cost vs Sales Analysis

Costs increase along with sales

In many cases, cost exceeds revenue

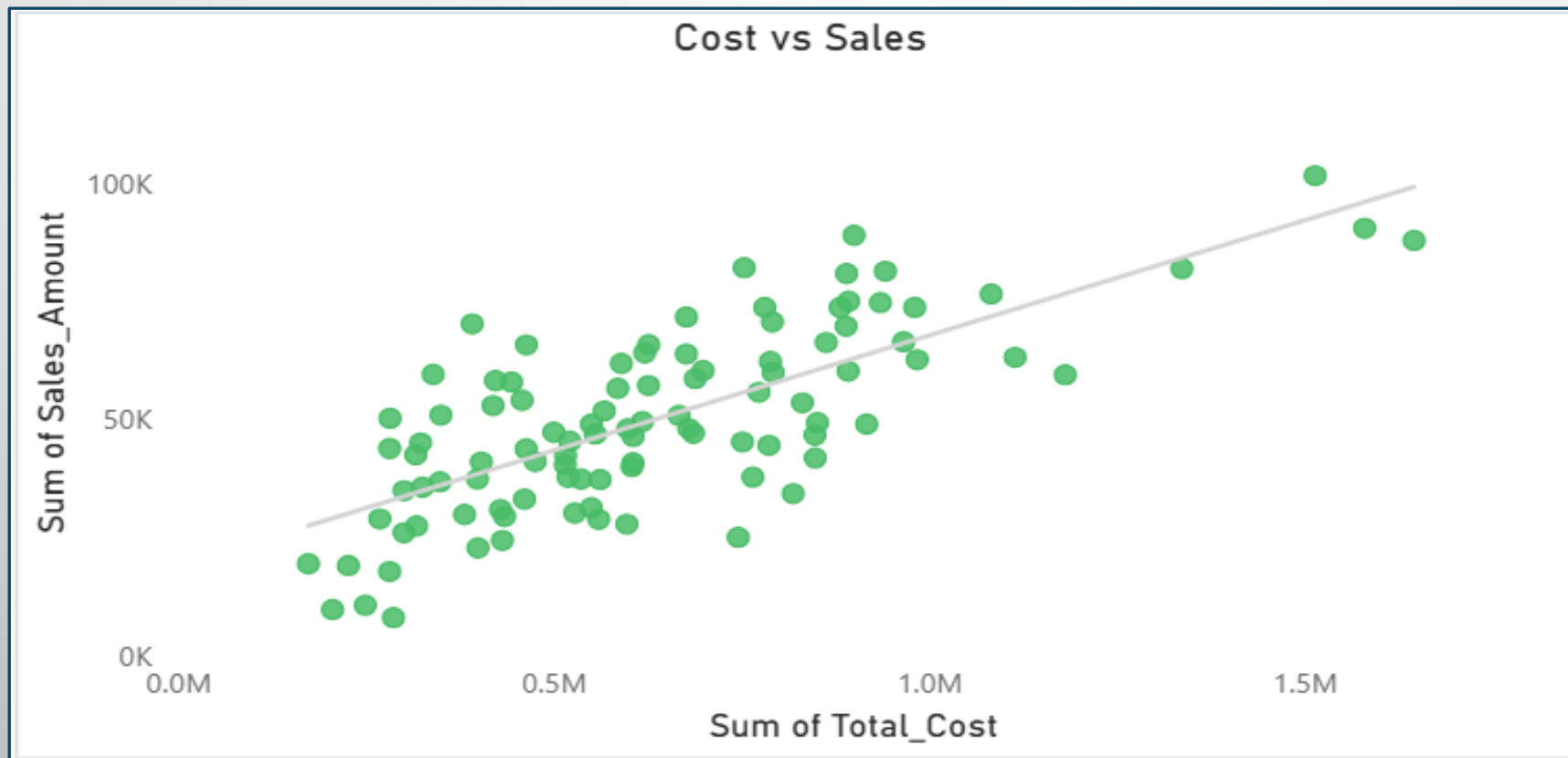
→ Leads to continuous losses



Points below = loss region

Key Insight

- High operational costs exceed revenue
→ Primary reason for business losses



Hypothesis Testing (Paired T-Test)

- H0: No difference between Sales and Cost
H1: Cost is higher than Sales

p-value < 0.05 → Significant result

T-statistic: -33.48637831556297

P-value: 1.9215896949403851e-165

Test Result

- T-statistic: -33.48
P-value: ~ 0

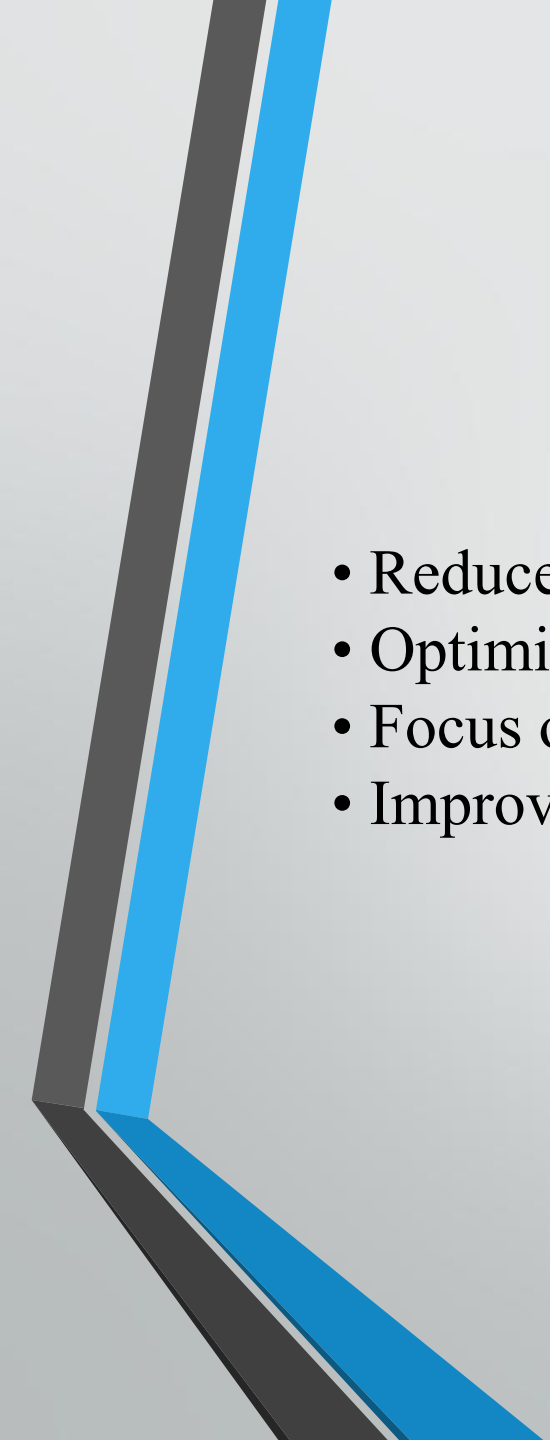
Conclusion:
Cost is significantly higher than sales

Conclusion

- High costs are the main reason for negative profitability.

The business is sales-driven but not profit-driven.

- This impacts long-term sustainability of the business.



Recommendations

- Reduce operational costs
- Optimize pricing strategy
- Focus on profit, not just sales
- Improve cost efficiency